Do diseases predict other diseases?

A multi-task learning deep neural network approach

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Research Question

Does the current prevalence of one chronic condition predict the prevalence of others, **above and beyond** non-invasive demographic features alone?

Motivation

Chronic disease comorbidity is the norm, not the exception. Predicting co-occurrence could support early screening without invasive tests.

Gap

Most prior work predicts each disease independently. Multi-task learning can model the joint structure of co-occurrence simultaneously.

Contribution

One MTL-DNN predicts 15 conditions jointly using disease flags as inputs, and quantifies how much they add over demographics alone.

Dataset: NHANES 2021-2023

Key figures

- $n = 7{,}269$ participants after cleaning
- 15 binary disease targets
- 17 demographic input features
- 70 / 15 / 15 train / val / test split

Demographic inputs (17)

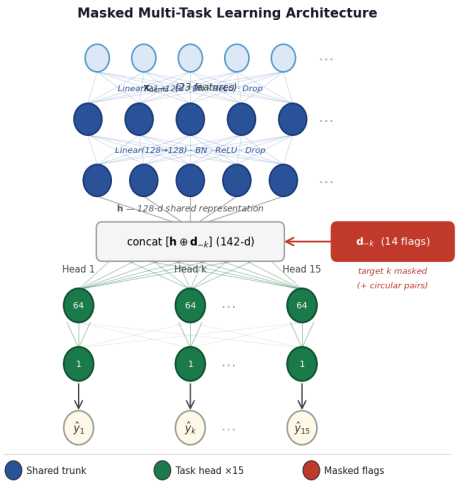
- Age, sex, BMI, waist circumference
- Systolic & diastolic BP (3-reading average)
- Smoking, alcohol, physical activity
- Hours worked, weight-loss attempt, insurance
- Self-rated health, mental health visits
- Race (5 dummies), salt intake (3 dummies)

Self-reported lifetime prevalence. NaN masking retains individuals with missing responses; absent flags imputed as 0 at inference.

Disease targets (15 binary flags)

Diabetes	Stroke
Pre-diabetes	Liver condition
Hypertension	Thyroid problem
High cholesterol	COPD/Emphysema
Arthritis	Cancer
Congestive HF	Kidney disease
Coronary HD	Asthma
Heart attack	

ML Method: Architecture



Shared trunk

17 demo inputs $\rightarrow$ FC(256) $\rightarrow$ FC(128) $\rightarrow$ 128-dim shared embedding

Task-specific heads ($K = 15$)

For each disease: embedding $\oplus$ 14 condition flags $\rightarrow$ FC(64) $\rightarrow \sigma$(output)

Circular masking

`CIRCULAR_PAIRS` prevents a disease appearing as its own input, removing trivial shortcuts

ML Method: Training and Baselines

Training protocol

- BCE loss with inverse-prevalence weighting
- Adam optimiser
- Dropout $= 0.3$
- Early stopping on validation PRAUC (patience $= 20$)
- Maximum 100 epochs

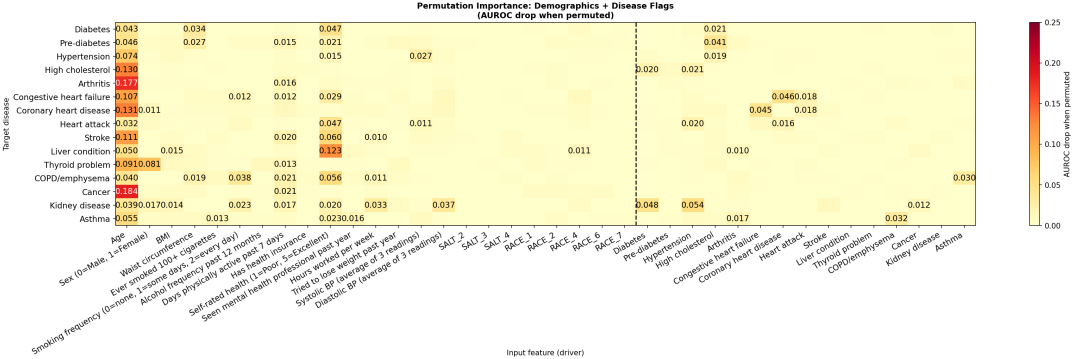
Why PRAUC for early stopping?

Prevalences range from 2.0% to 37.4%. PRAUC is more informative than AUROC under class imbalance and penalises poor recall on rare conditions more severely.

Four baselines

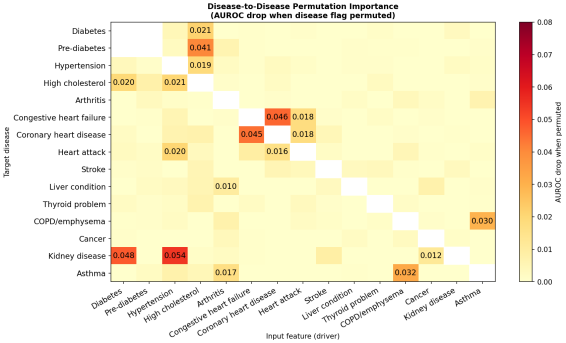
- **Logistic regression:** L-BFGS, one model per disease, all inputs, essentially our linear baseline
- **Random Forests** Split on Gini impurity, same as LR, one model per disease, all inputs, tells us if our non-linear interactions are important.
- **Single-task ANN:** One ANN trained independently per disease using all inputs, isolates the MTL architecture benefit
- **MTL-DNN (no disease inputs):** Identical architecture to MTL-DNN but disease flags are removed which isolates the contribution of co-occurrence information

Interpretability: Permutation Importance

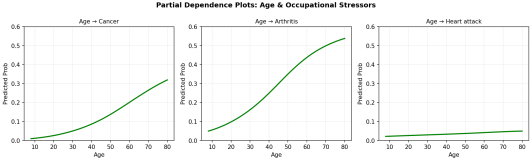


Interpretability: Disease-to-Disease Effects and PDPs

Disease-to-disease permutation importance

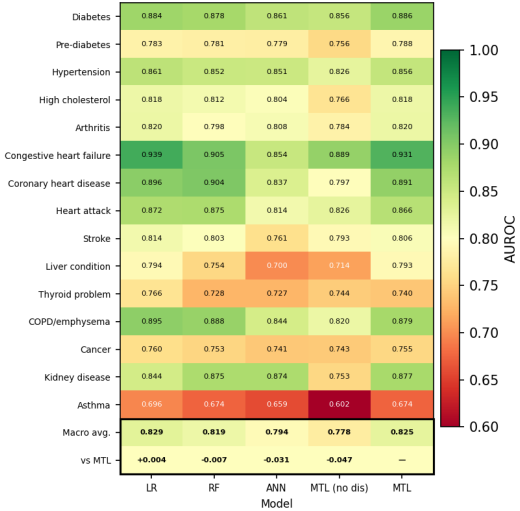


Partial dependence plots



- Emphysema raises predicted asthma probability by 10%
- Eight conditions each raise high-cholesterol risk by 3-17%, likely via shared dietary pathways not captured by the model
- Marginal effects held at population mean for all other inputs

Results: Model Comparison



Key figures

- Macro-avg AUROC: **0.822**
- **+5%** disease inputs over demo only
- **+3%** MTL over single-task ANNs
- MCC: 0.21 (stroke) to 0.56 (hypertension)

- **LR with disease inputs matches MTL-DNN**, suggesting co-occurrence is approximately linear at this sample size

- MTL gain consistent with Caruana (1997): shared trunk recovers comorbidity structure individual models cannot

Results: Benchmarks vs. Published Literature

Disease	Prev.	AUROC	Bench. AUROC	Δ
Diabetes	.103	.883	.830	+.053
Pre-diabetes	.116	.787	—	—
Hypertension	.346	.861	.770	+.091
High cholesterol	.374	.821	—	—
Arthritis	.315	.822	.814	+.008
Congestive HF	.029	.938	—	—
Coronary HD	.048	.895	.890†	+.005
Heart attack	.042	.840	—	—
Stroke	.042	.804	.753–.822†	±
Liver condition	.055	.778	—	—
Thyroid problem	.132	.745	—	—
COPD/emphysema	.068	.875	.900†	−.025
Cancer	.141	.754	—	—
Kidney disease	.020	.841	—	—
Asthma	.171	.684	.737†	−.053
Macro avg.	—	.822	—	—

Dashes: no comparable study found. † benchmark includes blood biomarkers.

- Meets or exceeds existing benchmarks, lacks only where lab biomarkers have been used
- Establishes new binary prediction benchmarks for 8 diseases in NHANES dataset
- Uses **fewer inputs** than most studies, which commonly include EHRs, DNA, and lab biomarkers
- Largest gains on hypertension (+9%) and diabetes (+5%)
- Only underperforms on COPD and asthma, both of which have benchmarks that include blood biomarkers

Challenges and How They Could Be Addressed

Limitation Self-reported lifetime prevalence

Diagnoses require both contracting and receiving a formal diagnosis, so under-reporting is systematic. Timing is also lost: a condition diagnosed yesterday is indistinguishable from one diagnosed 20 years ago.

Limitation NaN masking at inference

Absent disease flags are imputed as 0 at inference, risking bias towards "not present". A prevalence-weighted imputation or a separate missingness indicator would be more principled.

Limitation Linear ceiling

LR with disease inputs matches MTL-DNN, suggesting too much binary on binary prediction, NN doesn't have enough opportunity to learn non-linear interactions.
Also $n = 7{,}269$ may be too small to recover non-linear structure. Merging NHANES cycles from 1999 onwards would substantially increase power.

Future work

Add EHRs, lab biomarkers, or DNA when available; replace coarse categories such as "cancer" with site-specific flags to reveal associations currently masked by heterogeneity.

Summary

- Disease co-occurrence carries genuine predictive signal: adding disease flags lifts macro-average AUROC by **5 percentage points** over demographics alone.
- The MTL architecture contributes a further **3 percentage points** over separately trained ANNs, consistent with Caruana (1997), by learning a shared comorbidity representation.
- **The parity between logistic regression and MTL-DNN** is the most striking finding: disease co-occurrence appears approximately linear in this population and sample, raising the question of whether a neural network is strictly necessary here.
- The model meets or exceeds published NHANES benchmarks for the majority of conditions using only self-reported, non-invasive inputs.